



SIMATS ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCE
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



COURSE NAME: Theory of Computation

COURSE CODE: CSA13

COURSE OUTCOME COVERED

CO2: Design Finite Automata DFA, NFA, ϵ -NFA and demonstrate their equivalence for recognizing regular languages. (BL:3)

Assessment Tool Weightage: 40%

QUESTIONS: 5
Duration : 1 Hour

Total Marks:50

Design Task

Code of Conduct:

I Ebin Antony P (192373011) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

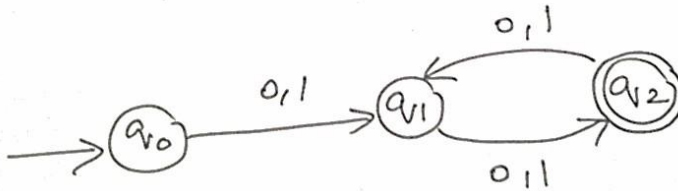
Signature of the student:



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1. Design the regular expression for the language represented by the given automata



2. Consider the following grammar

$S \rightarrow aSc | T | U | \epsilon$

$T \rightarrow bTc | \epsilon$

$U \rightarrow aUb | \epsilon$

A) What is the language defined by the grammar?

B) Give the derivation tree for the word aaabbc

C) A Grammar is Ambiguous if there exists two distinct derivations for a word. Is the above grammar ambiguous? Justify.

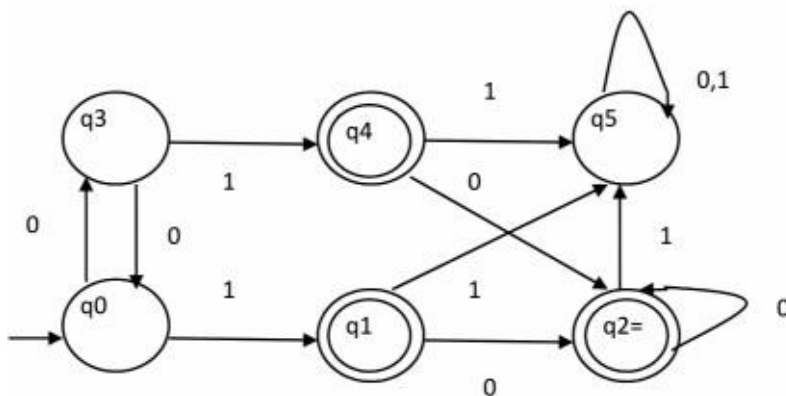
3. Design a non deterministic finite automata with ϵ -transitions to represent the language denoted by the regular expression

$00^*1 + 11^*0$

4. i) Show that if L_1 and L_2 are regular then $(L_1 + L_2)$ also regular

ii) Is the language $\{a^p \mid p \text{ is prime}\}$ regular? Justify

5. Design minimized automata, Consider the DFA given below. Find the states that are equivalent and construct a minimum state automata equivalent to the given automata. Draw the minimized automata, write its transition table and mark initial and final states



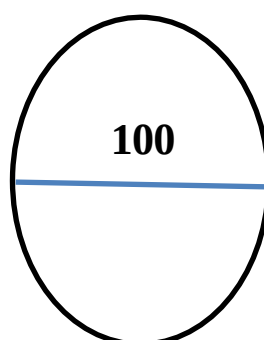


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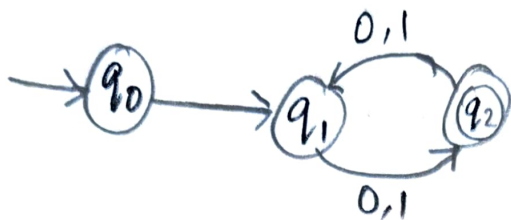


RUBRICS FOR CALCULATION

Criteria	Weight age	Q1 (10)	Q2 (10)	Q3 (10)	Q4 (10)	Q5 (10)	Total Marks (50)
Conceptual Understanding	30	3	3	3	3	3	15
Accuracy of Answer	25	2.5	2.5	2.5	2.5	2.5	12.5
Application of Knowledge	20	2	2	2	2	2	10
Completeness of Response	15	1.5	1.5	1.5	1.5	1.5	7.5
Clarity & Presentation	10	1	1	1	1	1	5
Total per Question	10 Marks	/10	/10	/10	/10	/10	/ 50



① 1. Regular Expression for the Given Automation



From the automation:

- q_0 is the initial state.
- From q_0 , on 0 or 1, go to q_1 .
- From q_1 , on 0 or 1, go to final state q_2 .
- From q_2 , on 0 or 1, return to q_1 .
- q_2 is the final state.

Let: $\Sigma = (0+1)$

The regular expression is:

$$(0+1)(0+1)((0+1)(0+1))^*$$

$$\Rightarrow \underline{\underline{\{(0+1)^2((0+1)^2)^*\}}}$$

Examples accepted: ...

00, 01, 10, 11, 0000, 0101, 1010, etc...

2. Given Grammar

$$S \rightarrow aSc | T | U | \epsilon$$

$$T \rightarrow bTc | \epsilon$$

$$U \rightarrow aUb | \epsilon$$

A) Language defined by the grammar

From T: $T \rightarrow bTc | \epsilon$

$$\text{So, } L(T) = \{b^n c^n \mid n \geq 0\}$$

From U: $U \rightarrow aUb | \epsilon$

$$\text{So, } L(U) = \{a^n b^n \mid n \geq 0\}$$

From S: $S \rightarrow aSc$ adds the same no. of a's at the beginning and c's at the end.

$$\therefore L = \{a^i b^j c^{i+j} \mid i, j \geq 0\} \cup \{a^i b^j c^i \mid i, j \geq 0\}$$

B) Derivation tree for aaabbc

we need: aaabbc

Use $S \rightarrow aSc$ once: $S \Rightarrow aSc$

Then $S \Rightarrow aUb$

So $aSc \Rightarrow aaUbc$

Again $U \Rightarrow aUb$

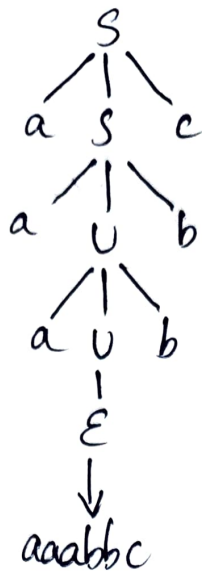
Therefore $aaUbc \Rightarrow aaaUbbc$

Finally $U \Rightarrow \epsilon$

Thus :

$$\boxed{S \Rightarrow aSc \Rightarrow aaUbc \Rightarrow aaaUbbc \Rightarrow aaabbc}$$

Derivation Tree.



c) Yes, the grammar is ambiguous.

Derivation 1: $S \Rightarrow aSc \Rightarrow ac$ (Inner S becomes ϵ)

Derivation 2: $S \Rightarrow aSc$. (inner S can become $T: \Rightarrow aTc$

Then: $T \Rightarrow \epsilon$

$\therefore \text{~~acac~~} \Rightarrow ac$

So there is two different derivations:

$$S \rightarrow aSc \rightarrow ac$$

ϵ

$$S \rightarrow aSc \rightarrow aTc \rightarrow ac$$

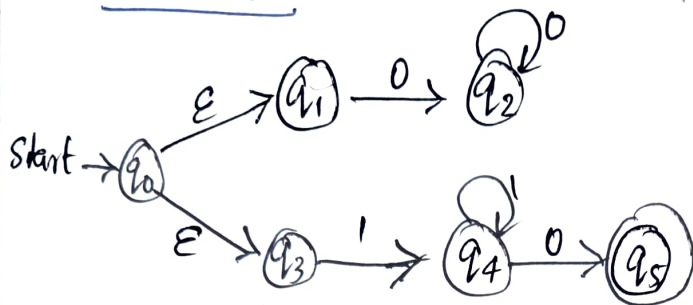
Hence, The grammar is ambiguous.

3. NFA with ϵ -transitions.

Given RE : $00^* + 11^*0$

Two alternatives are 00^* and 11^*0

NFA Construction



Transition Table

State	0	1	ϵ
q_0	—	—	q_1, q_3
q_1	q_2	—	—
q_2	q_2	—	—
q_3	—	q_4	—
q_4	—	q_4	—
q_5	—	—	—

1st part:

$q_1 \xrightarrow{0} q_2, q_2 \xrightarrow{0} q_2$ gives: 00^*

2nd part:

$q_3 \xrightarrow{1} q_4, q_4 \xrightarrow{1} q_4, q_4 \xrightarrow{0} q_5$ gives: 11^*0

Hence NFA represents:

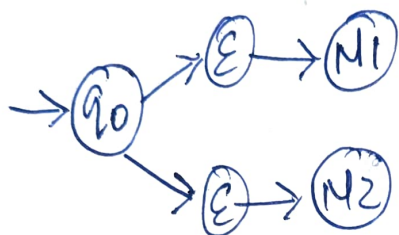
$00^* + 11^*0$

4. Regular languages.

1) Prove that if L_1 and L_2 are regular, then $L_1 \cup L_2$ is also regular.

Given L_1, L_2 are regular languages.

$\therefore M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1), M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$
which accept L_1 and L_2 .



Final states are: $F = F_1 \cup F_2$

$\therefore L_1 \cup L_2$ is regular

Hence, regular languages are closed under union.

11) $\{a^p \mid p \text{ is prime}\}$ regular?

The language is not regular.

Justification using Pumping lemma

Assume that L is regular.

Then by pumping lemma, there exist a pumping length n .

Choose a prime number p such that: $p > n$

Consider: $w = a^p$

Since, p is prime: $w \in L$

By the Pumping Lemma: $w = xyz$

where: $|xy| \leq n$ and $|y| > 0$

Since the strings only contains a's: $y = a^k$ ie, $k \geq 1$

pump y $p+1$ times : $xy^{p+1}z$

Its length becomes: $p+pk = p(1+k)$

This is a composite number, because both factors are greater than 1.

$$\therefore xy^{p+1}z \notin L$$

This contradicts the Pumping Lemma

Hence: $L = \{a^p \mid p \text{ is prime}\}$ is not regular.

5. DFA Minimization

From the given DFA:

- Initial state = q_0 , Final states = q_1, q_2, q_4
- Non-final states = q_0, q_3, q_5

Transition Table

State	0	1	Final ?
q_0	q_3	q_1	No
q_1	q_2	q_5	Yes
q_2	q_2	q_5	Yes
q_3	q_0	q_4	No
q_4	q_2	q_5	Yes
q_5	q_5	q_5	No

Step 1: Initial Partition

Separate final & non-final states:

$$P_0 = \{\{q_1, q_2, q_4\}, \{q_0, q_3, q_5\}\}$$

Step 2: Check equivalent states

q_1, q_2, q_4

For q_1 :

- $0 \rightarrow q_2$ (final group)
- $1 \rightarrow q_5$ (non-final group)

For q_2 :

- $0 \rightarrow q_2$ (final group)
- $1 \rightarrow q_5$ (non-final group)

For q_4 :

- $0 \rightarrow q_2$ (final group)
- $1 \rightarrow q_5$ (non-final group)

All three have identical behaviour

$$\therefore \boxed{q_1 = q_2 = q_4}$$

q_0 and q_3

For q_0 :

- $0 \rightarrow q_3$
- $1 \rightarrow q_1$

For q_3 :

- $0 \rightarrow q_0$
- $1 \rightarrow q_4$

Since q_0 and q_3 both transition to the same groups

$$\boxed{q_0 = q_3}$$

q_5 remains separate

q_5 is not equivalent to q_0/q_3 bcoz q_5 loops on both inputs

Equivalent States

∴ Minimized DFA has 3 states

$\{q_0, q_3\}, \{q_1, q_2, q_4\}, \{q_5\}$

- $A = \{q_0, q_3\} \rightarrow \text{initial}$
- $C = \{q_5\} \rightarrow \text{non-final}$

• $B = \{q_1, q_2, q_4\} \rightarrow \text{final}$

Minimized Transition Table

Minimized State	0	1
$A = \{q_0, q_3\}$	A	B
$B = \{q_1, q_2, q_4\}$	B	C
$C = \{q_5\}$	C	C

Initial State : A

Final State : B

DFA diagram (Minimized)

